Notes 4: The Linux FS

Each of the commands used for navigating the file system

> Here is an example from the previous notes! Use the same for mat for the commands to navigate the file system!

Echo

Definition:

- Used for displaying text on the screen.*

Usage:

- `echo` + `option` + `string to print`

Examples:

- Display/print a line of text
  - `echo "hello world"`
- Display a line of text with a horizontal tab
  - `echo -e "\thello world"`
- Display 2 lines of text with a single echo command
  - `echo -e "Line 1\nLine2"`
- Display 2 lines of text with a single echo command, with the second line starting with a tab
  - `echo -e "Line 1\n\tLine2"`
- Display 2 lines of text with a single echo command that starts with a tab
  - `echo -e "\tLine 1\tLine2"`

Definitions of the following terms:

- **File system**
  - The way files are stored and organized
- **pathname**
  - In a filesystem every file has a **pathname** which indicates the loction of the file in the filesystem (like an address)
- **Absolute path**
  - The location of a file starting a the root of the file system
- **Relative path**
  - The loction of a file starting from the current working directory or a dirctory that is located inside the current working directory
- **The difference between your home directory and the home directory**
  - Your home directory is your personal space where all your files are stored and managed (e.g., /home/maria53), while the home directory (/home) is the parent directory that contains all users' home directories on the system.
- **parent directory**

  - A dirtectory containing one or more directories and files.
- **child directory or subdirectory**
  - A better name for this is a subdirectory or subfolder. This is a directory inside another directory. See image for visual reference.
- **Bash special characters**
  - . (single period)
  - .. (2 consecutive periods)
  - ~ (tilde character)
  - / (one forward slash)
- **environment variables**
  - Are used by the small to track specific system information and user information.
- **user defined variables**
  - In programming, a variable is place to store data. A variable is like a box with a label
- **Why do we need use $ with variables in bash shell scripting?**
  - To use the value stored in an environment variable you must prepend the variable name with a $. Here are some useful environment variables:
  - $USER = stores the current's user username
  - $HOME = stores the absolute path of current's user home directory
  - $PWD = stores the absolute path of the present working directory.
  - $OLDPWD = stores the absolute path of the previous current working directory.